

VLSI Design and Testing(BEC602)

CMOS Circuit and Logic Design

1 Introduction

In earlier discussions, we introduced CMOS logic under the assumption that MOS transistors behave as ideal switches.

However, real MOS transistors exhibit certain limitations that deviate from this idealized behavior.

So far, we have primarily focused on fully complementary CMOS logic structures and the ratioed CMOS inverter.

Here we will explore:

1. **Alternative CMOS logic configurations** beyond fully complementary logic.
2. **The impact of non-ideal switch behavior** on circuit performance.
3. **A comparative analysis of different logic structures** to evaluate their advantages and trade-offs.

1.1 Key Design Considerations

When designing CMOS circuits, two crucial aspects must be addressed to achieve the desired performance:

1. **Circuit (Structural) Design** – This involves selecting the appropriate logic configuration and transistor arrangements.
2. **Layout (Physical) Design** – This deals with the physical placement and interconnections of transistors on a chip.

These two design phases are closely interconnected.

The behavior of circuits at a lower level can influence high-level architectural decisions.

Therefore, system designers must have a fundamental understanding of low-level circuit design to make informed choices at higher levels of abstraction.

2 CMOS Logic Structures

In certain situations, fully complementary static CMOS gates may not be the most efficient choice.

The area required for implementation might be excessive, the speed may be too slow, or the function may not be directly realizable using a purely complementary structure — such as in the case of large Programmable Logic Arrays (PLAs).

In such cases, alternative CMOS logic structures can be employed to design smaller and faster gates.

However, these improvements often come at the cost of increased design complexity, greater operational challenges, and potential reductions in circuit stability.

2.1 Need for Alternative CMOS Logic Structures

The motivation for exploring alternative CMOS logic structures arises from the following factors:

Area Constraints: Fully complementary CMOS gates can occupy a large silicon area, which is undesirable in high-density designs.

Speed Limitations: The propagation delay of traditional CMOS logic may be too high for certain high-performance applications.

Design Feasibility: Some logic functions, such as those required in large PLAs, may not be efficiently implemented using complementary CMOS alone.

To address these issues, alternative logic styles are explored, each with its own trade-offs in terms of power consumption, speed, and complexity.

3 CMOS Complementary Logic

Complementary CMOS logic is widely used due to its robustness and ratioless operation.

3.1 Properties of CMOS Complementary Logic

Ratioless Operation: All complementary CMOS gates can be designed such that the circuit functions correctly even when all transistors are of equal size.

Optimization: Although basic functionality is achieved with uniform transistor sizing, speed optimization techniques involving different transistor sizes will be introduced later.

Complex Logic Implementation: More complex gates can also be implemented using CMOS logic.

For example, a complex gate implementing the function:

$$Z = (A \cdot B) + C \cdot (D + E)$$

is used as a basis for comparison between various logic families.

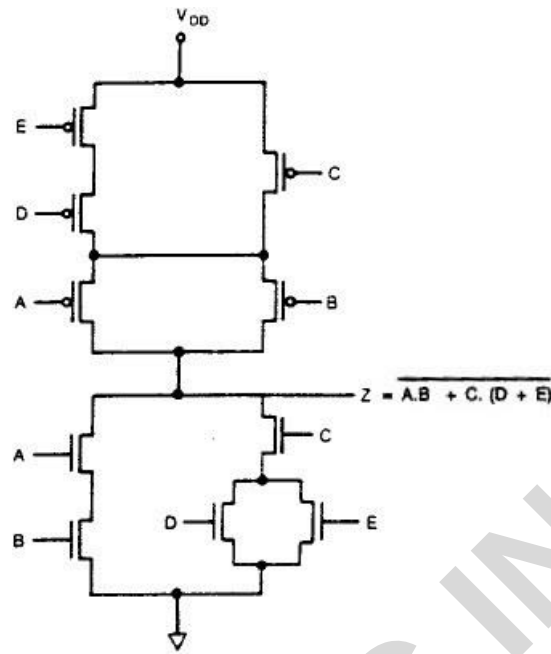


Figure 1: CMOS Complementary Logic

3.2 Substrate Connection Considerations

In the schematics, the substrate connection may be omitted. But it plays a crucial role in MOS circuit behavior. The substrate connection impacts key electrical properties, such as:

Body Effect: Affects the threshold voltage of transistors, influencing circuit performance.

Leakage Currents: Improper substrate connections may lead to increased leakage or latch-up issues.

Power Integrity: Proper substrate biasing ensures reliable operation across varying process conditions.

This understanding of complementary CMOS logic forms the foundation for further discussions on optimizing logic circuits for speed and efficiency.

4 Pseudo - nMOS Logic

A typical pseudo - nMOS gate is illustrated in figure below.

In this configuration, a single pMOS transistor acts as the load device, with its gate permanently connected to ground (V_{SS}).

This design is similar to conventional nMOS logic, where the depletion or enhancement-mode nMOS load transistor is replaced with a pMOS device.

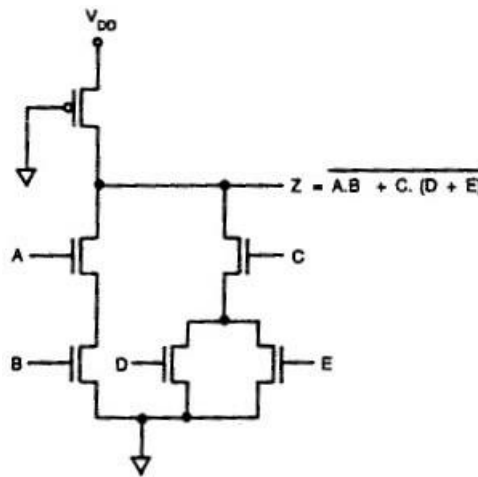


Figure 2: Pseudo - nMOS Logic

4.1 Operation and Design Considerations

The pull-up device (pMOS) is always turned on, meaning the circuit operates with **rationed logic**, where the gain ratio of the p-load to the nMOS driver transistors, $\beta_{load}/\beta_{driver}$, must be carefully selected.

The transistor sizing must ensure proper switching and logic level generation. The effective β_n/β_p ratio must be chosen in accordance with values derived from circuit equations.

One of the main drawbacks of pseudo-nMOS logic, as with conventional nMOS logic, is **static power dissipation**. Since the pMOS load is always on, current continuously flows whenever the pull-down network is active.

4.2 Transistor Count and Input Loading

An n -input pseudo-nMOS gate requires **$n+1$ transistors**.

In a fully complementary CMOS gate, the capacitive load on each input is at least two unit gate loads. In pseudo-nMOS logic, the minimum input load can be just one unit gate load since only a single transistor is used per input term.

However, if minimum-sized driver transistors are used, the pull-up gain must be reduced to maintain adequate noise margins, which in turn slows the gate's rise time.

4.3 Comparison with Conventional nMOS and CMOS Logic

Pseudo-nMOS gates do not have a significant advantage over conventional nMOS depletion load gates, except that they allow **emulation of nMOS circuits in a CMOS process**.

One potential advantage of using a pMOS load is the **absence of the body effect**, which affects nMOS depletion loads.

A pseudo-nMOS gate can offer a **higher circuit density** compared to fully complementary CMOS gates due to the reduced transistor count.

In summary, pseudo-nMOS logic provides a method for implementing logic gates with fewer transistors, but at the cost of static power dissipation and slower rise times in some cases.

5 Dynamic CMOS Logic

Dynamic CMOS logic is a logic family that reduces transistor count while achieving high-speed operation.

Unlike static CMOS logic, which maintains a stable output using both pull-up and pull-down networks, dynamic CMOS logic relies on clocked operation and charge storage principles.

The main features of dynamic CMOS logic are:

- Uses clock-driven precharge and evaluation phases.
- Reduces transistor count, leading to increased logic density.
- Offers higher speed compared to static CMOS due to reduced capacitance.
- Requires careful clock synchronization to avoid errors.

5.1 Structure and Working of Dynamic CMOS Gates

5.1.1 Basic Structure

A simple dynamic CMOS logic gate is shown in figure below.

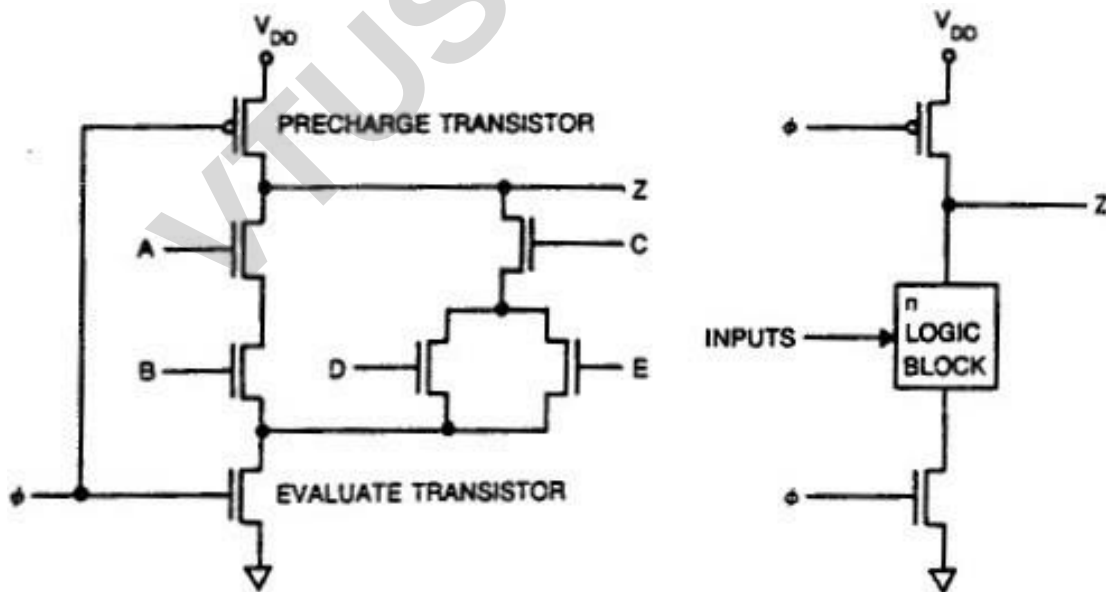


Figure 3: Dynamic CMOS Logic

It consists of:

An nMOS logic network, responsible for evaluating the logic function.

A pMOS precharge transistor, which charges the output to V_{DD} during the precharge phase.

An nMOS evaluation transistor, which selectively discharges the output during evaluation.

A clock signal (ϕ), controlling the switching between precharge and evaluation phases.

5.1.2 Operation Phases

The dynamic CMOS gate operates in two main phases, controlled by the clock signal ϕ :

1. Precharge Phase ($\phi = 0$)

The pMOS precharge transistor turns ON, charging the output node to V_{DD} .

The nMOS evaluation transistor is OFF, preventing any discharge.

The output is temporarily set to HIGH regardless of input values.

2. Evaluation Phase ($\phi = 1$)

The pMOS precharge transistor turns OFF.

The nMOS evaluation transistor turns ON, allowing the nMOS logic block to evaluate the function.

If the input logic network forms a conducting path to ground, the output discharges to LOW.

If no conducting path exists, the output remains at its precharged HIGH state.

This approach reduces the number of transistors needed, since a pull-up network is not required.

5.2 Limitations and Challenges

Although dynamic CMOS logic provides higher speed and reduced area, it has several challenges:

5.2.1 Charge Leakage and Charge Sharing

The output node holds charge dynamically without a steady pull-up.

Over time, leakage currents can reduce the output voltage, causing incorrect logic levels.

Charge sharing between nodes can alter the stored charge, leading to unstable outputs.

5.2.2 Input Switching Restriction

Inputs must only change during the precharge phase.

If inputs change during evaluation, charge redistribution effects may corrupt the output.

5.2.3 Cascading Issue

Dynamic CMOS gates cannot be directly cascaded using a single-phase clock.

The first gate must evaluate before the next stage begins evaluation, causing timing constraints.

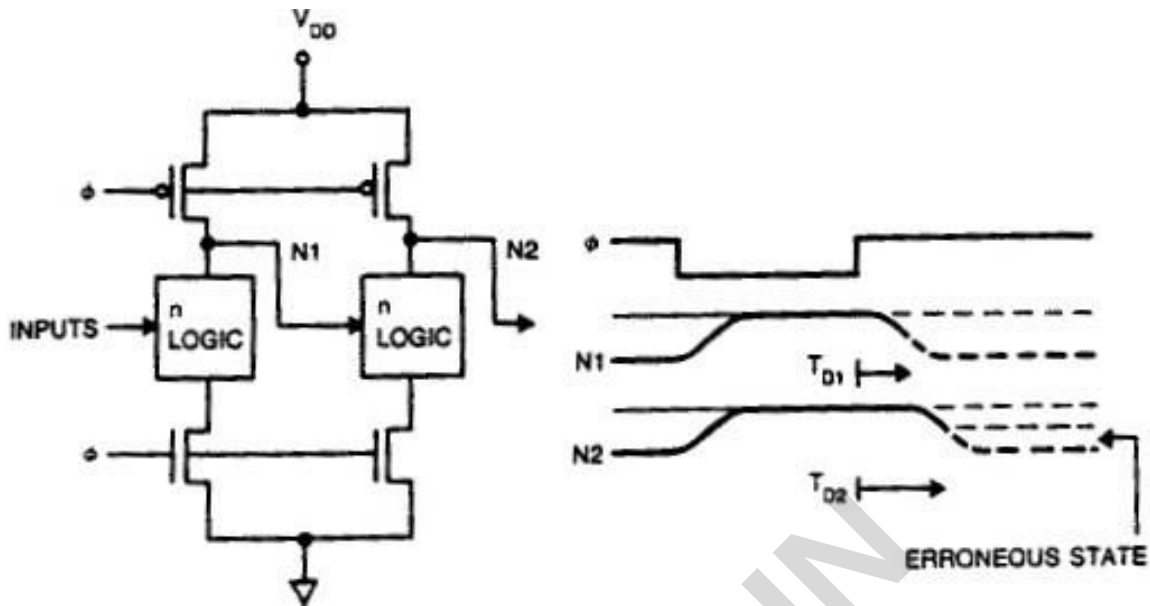


Figure 4: Cascaded Dynamic CMOS

5.3 Multi-Phase Clocking Strategies

To overcome cascading issues, multi-phase clocking techniques are used. These techniques introduce additional clock phases that improve charge retention, control signal propagation, and enable correct logic evaluation.

5.3.1 Four-Phase Dynamic Logic

The four-phase logic approach introduces a structured clocking sequence that ensures that outputs are properly evaluated and stored before being used as inputs in the next stage.

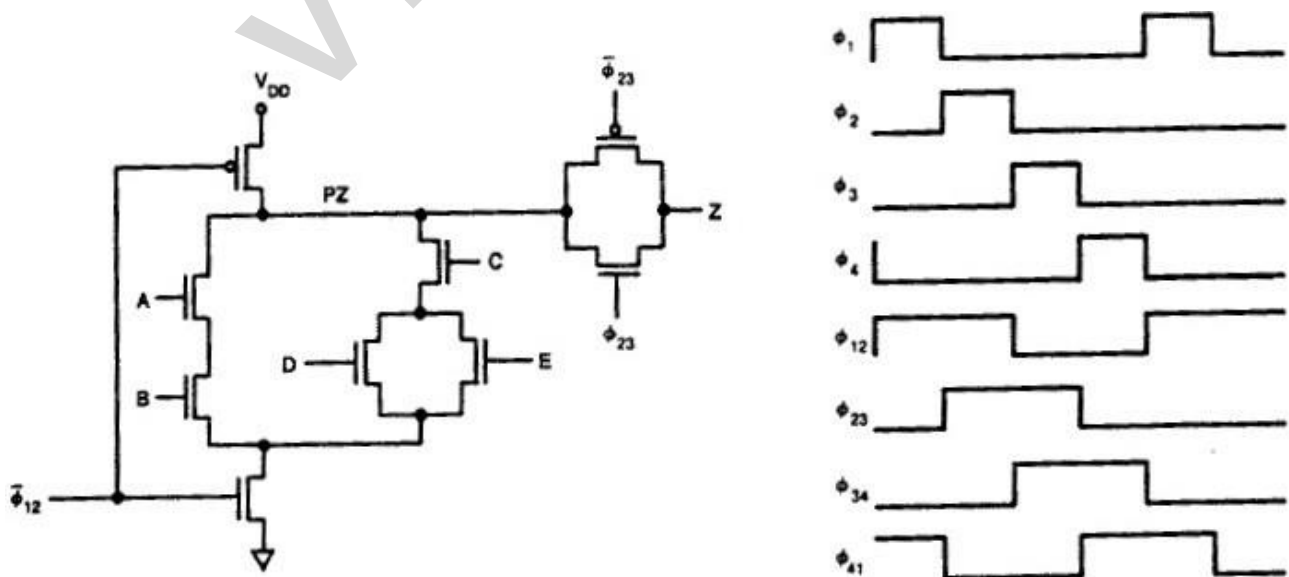


Figure 5: 4- Phase Logic

Working Principle The four-phase logic cycle consists of:

Phase ϕ_1 (Precharge of PZ): During this phase, node PZ is precharged to V_{DD} , while node Z retains its previous value.

Phase ϕ_2 (Precharge of Z): The precharge of node PZ is maintained, and a transmission gate turns on, allowing node Z to also precharge.

Phase ϕ_3 (Evaluation): The gate evaluates, and if the pull-down network is activated, node PZ discharges conditionally.

Phase ϕ_4 (Hold Phase): Node Z is held in its evaluated state, ensuring that it does not change prematurely.

Advantages

Reduces charge-sharing issues by ensuring proper storage of values before evaluation.

Enables cascading of multiple logic stages without the risk of premature discharge.

Ensures that evaluated values remain stable during computation.

Limitations

Requires four separate clock signals, increasing circuit complexity.

Additional clocking transistors increase power consumption.

Strict timing requirements must be met for reliable operation.

5.3.2 Two-Phase Dynamic Logic

An alternative to four-phase logic is a two-phase approach, which simplifies clocking at the cost of slightly reduced reliability.

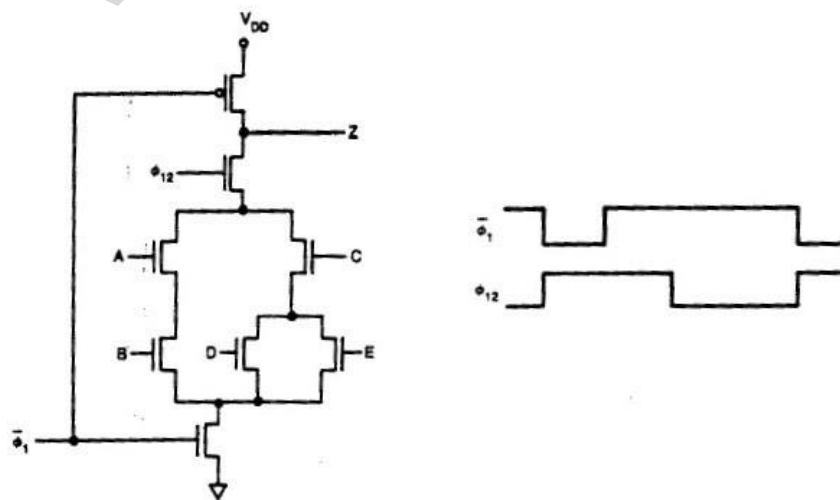


Figure 6: 2- Phase Logic

Working Principle A two-phase logic scheme utilizes alternating clock cycles:

Phase ϕ_A : Certain gates evaluate their logic function.

Phase ϕ_B : Other gates evaluate, while the outputs of the previous phase are held constant.

Advantages

Reduces the number of required clock signals from four to two.

Simpler circuit design and reduced power consumption compared to four-phase logic.

Limitations

More susceptible to charge leakage compared to four-phase logic.

Requires careful synchronization to prevent timing mismatches.

Table 1: Comparison of Two-Phase and Four-Phase Logic

Feature	Four-Phase Logic	Two-Phase Logic
Clock Signals	Four ($\phi_1, \phi_2, \phi_3, \phi_4$)	Two (ϕ_A, ϕ_B)
Stability	Higher, due to hold phases	Moderate, more charge leakage
Circuit Complexity	Higher	Lower
Power Consumption	Higher	Lower
Suitability for Cascading	More suitable	Requires careful synchronization

5.4 Transistor Count Analysis

The number of transistors in dynamic CMOS logic is significantly lower than in static CMOS:

A static CMOS logic gate requires $2N$ transistors for an N -input gate.

A dynamic CMOS gate requires $N+2$ transistors.

The reduced transistor count results in lower capacitance and higher speed.

5.5 Advantages of Dynamic CMOS Logic

High Speed: Reduced capacitance leads to faster switching times.

Lower Power Consumption: Only consumes dynamic power during switching.

Reduced Area: Uses fewer transistors compared to static CMOS.

5.6 Disadvantages of Dynamic CMOS Logic

Charge Leakage: Output logic levels degrade over time.

Timing Constraints: Requires careful synchronization of clock signals.

Complex Cascading: Cannot directly cascade without multi-phase clocking.

A standard CMOS logic gate.

Additional transistors controlled by a clock signal (ϕ) to regulate switching behavior.

During operation:

When $\phi = 0$, the circuit holds its previous state.

When $\phi = 1$, normal evaluation occurs.

6.2 Advantages of C MOS Logic

Reduced Power Dissipation: Originally designed for metal-gate CMOS, the clocked design minimizes unnecessary switching activity.

Latching Capability: C MOS logic inherently functions as a latch, holding data when the clock is low.

Compatibility with Dynamic Logic: It effectively interfaces with other dynamic logic families, improving timing synchronization.

6.3 Limitations of C MOS Logic

Increased Rise and Fall Times: Due to additional clocking transistors in series, switching speed is reduced.

Higher Input Capacitance: Although similar to static CMOS gates, the presence of clocking elements increases parasitic capacitance.

Complex Timing Requirements: The circuit behavior depends on the correct synchronization of clock signals.

6.4 Applications

C MOS logic is widely used in:

Low-power digital circuits.

Clocked latches and flip-flops.

Interface circuits between static and dynamic logic.

6.5 Conclusion

Clocked CMOS logic provides a power-efficient and synchronous alternative to conventional static CMOS gates. While it introduces increased rise and fall times, its benefits in latch-based designs and dynamic logic interfacing make it a crucial component in modern VLSI systems.

7 Cascade Voltage Switch Logic (CVSL)

Cascade Voltage Switch Logic (CVSL) is a differential CMOS logic style that utilizes both true and complement input signals for computation.

Unlike conventional static CMOS logic, CVSL relies on two complementary nMOS switching networks, combined with cross-coupled pMOS transistors for positive feedback.

7.1 Structure and Operation

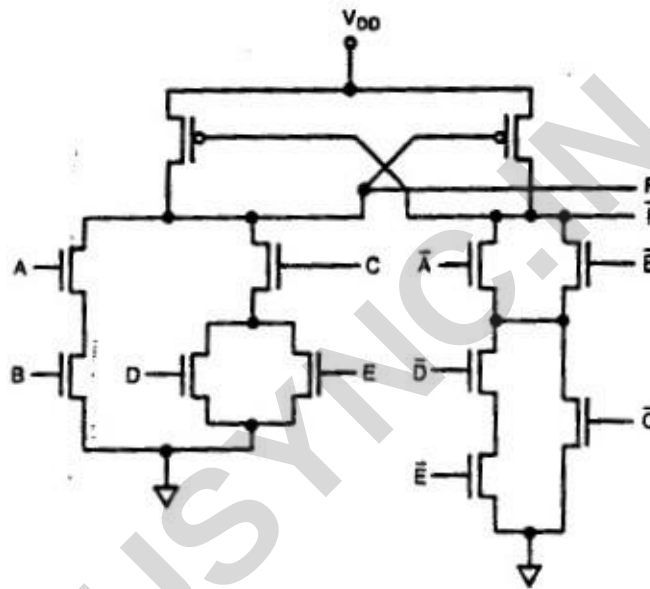


Figure 8: Cascade Voltage Switch Logic (CVSL)

A typical CVSL gate consists of:

Two complementary nMOS logic trees that implement the desired logic function.

Cross-coupled pMOS pull-up transistors that ensure positive feedback and bistable operation.

The key working principle involves:

1. **Differential Signaling:** The logic trees receive both the input and its complement, ensuring robust operation.
2. **Pull-Up Mechanism:** The cross-coupled pMOS transistors ensure that one output node is strongly pulled high while the other remains low.
3. **Switching Mechanism:** When the inputs change, one of the nMOS logic trees will conduct, forcing one node to be pulled low while the other remains high, resulting in a strong logic transition.

7.2 Advantages of CVSL

Strong Signal Integrity: Positive feedback ensures sharp transitions and improved noise immunity.

Logical Completeness: CVSL can implement any logic function efficiently, making it suitable for automated logic synthesis.

Lower Power Consumption: Unlike domino logic, CVSL gates do not require precharge and evaluate cycles, reducing dynamic power dissipation.

7.3 Limitations of CVSL

Increased Complexity: CVSL requires both true and complement signals, increasing the number of interconnections.

Larger Area Requirement: Additional routing and double-rail logic consume more chip area.

Slower Switching Speed: During transitions, the pMOS pull-ups must counteract the nMOS pull-down networks, leading to delays compared to conventional static CMOS gates.

7.4 Applications

High-speed arithmetic and logic units (ALUs) in microprocessors.

Circuits requiring differential signaling for noise reduction.

Automated logic synthesis applications where arbitrary logic expressions need to be implemented efficiently.

7.5 Conclusion

CVSL is a powerful logic family that provides robust differential operation and logical completeness at the cost of increased complexity and area. While it may not always be as fast as traditional CMOS or domino logic, its ability to implement any logic function makes it useful in specific applications.

8 Pass Transistor Logic

Pass Transistor Logic (PTL) is an alternative logic design methodology that reduces transistor count and enhances circuit efficiency.

It is particularly popular in **nMOS circuits**, with a fundamental application being the **2-input multiplexer**.

A notable implementation of PTL is found in the **ALU function unit of the OM-1 computer**.

8.1 nMOS and CMOS Implementations of Pass Transistor Logic

8.1.1 nMOS Pass Transistor Logic

In **nMOS-based PTL**, the pass transistors are used to steer signals to the output based on control inputs.

Figure below illustrates the nMOS structure, where only **nMOS transistors** are used.

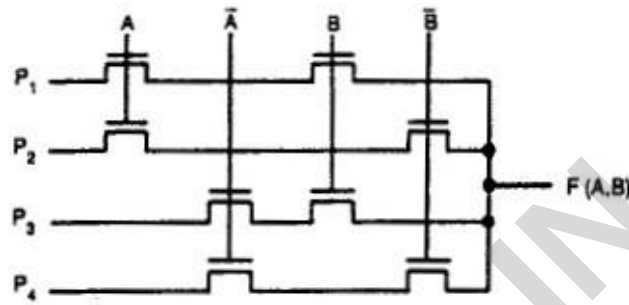


Figure 9: nMOS PTL

Advantages:

Provides the **fastest fall time** due to strong nMOS pull-down.

Disadvantages:

Suffers from **threshold voltage loss** (V_{th} drop), leading to degraded high-level signals.

8.1.2 CMOS Pass Transistor Logic

To overcome the limitations of nMOS PTL, a **CMOS version** can be designed by replacing each nMOS transistor with a **full transmission gate**, as shown in figure below.

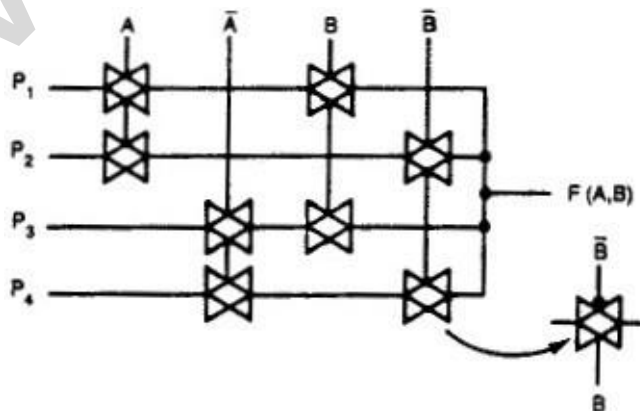


Figure 10: CMOS PTL

CMOS PTL uses both **nMOS** and **pMOS** transistors, ensuring full voltage swing.

This implementation provides **strong pull-up and pull-down** characteristics.

8.2 Alternative Implementations of Pass Transistor Logic

8.2.1 Optimized Layout for CMOS PTL

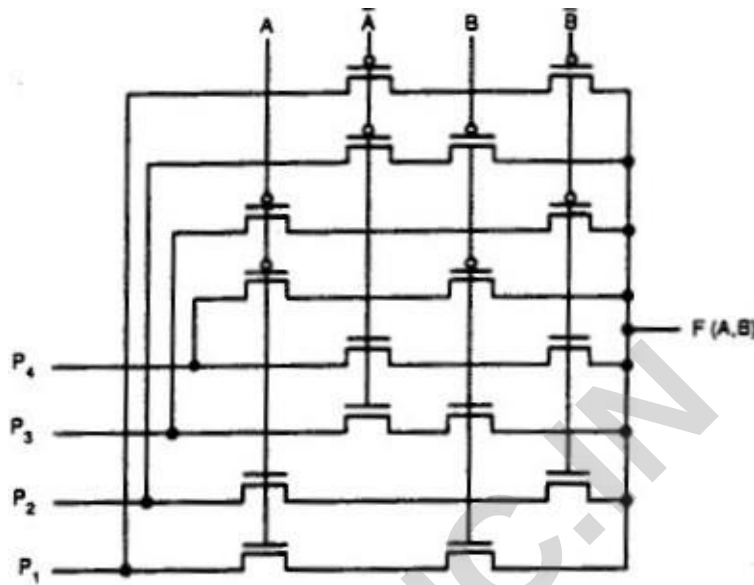


Figure 11: Modified CMOS for better layout

A more realizable layout is depicted in above figure, which:

Reduces direct **n-to-p transistor connections**.

Enhances circuit **practicality and performance**.

8.2.2 Dynamic Pass Transistor Logic

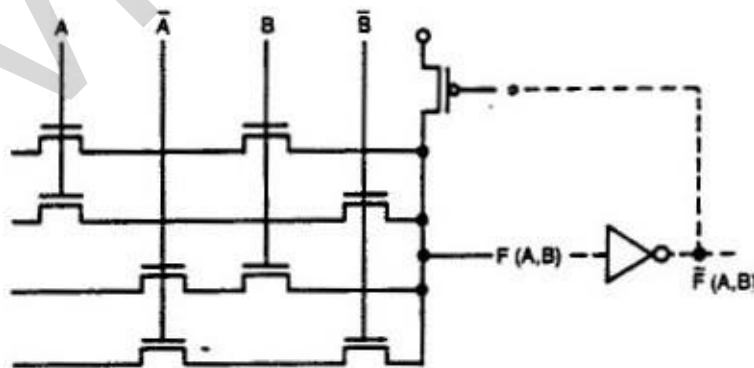


Figure 12: Dynamic PTL (p-pullup version)

Figure above shows a **dynamic version** of PTL, which requires a **precharge phase** before operation.

Performance comparison:

Comparable speed to **nMOS PTL**.

Requires a **precharge period**, which may **extend clock cycle times**.

8.2.3 Static Pass Transistor Logic with Feedback Buffer

An alternative to the dynamic version is the **static PTL implementation**, where:

A **feedback buffer** drives the **p-transistor pull-up**.

This design achieves **zero DC power dissipation**.

To ensure correct operation, the **p-transistor pull-up** and **n-transistor pull-down** must be **properly ratioed**.

8.3 Formal Methods for Designing Pass Transistor Logic

Formal methods for deriving pass transistor networks in **nMOS** have been developed by Whitaker. These methods are based on:

1. A **model** as shown in figure below, where control variables steer a pass transistor network.
2. The **pass function**, which determines how input variables propagate through the network.

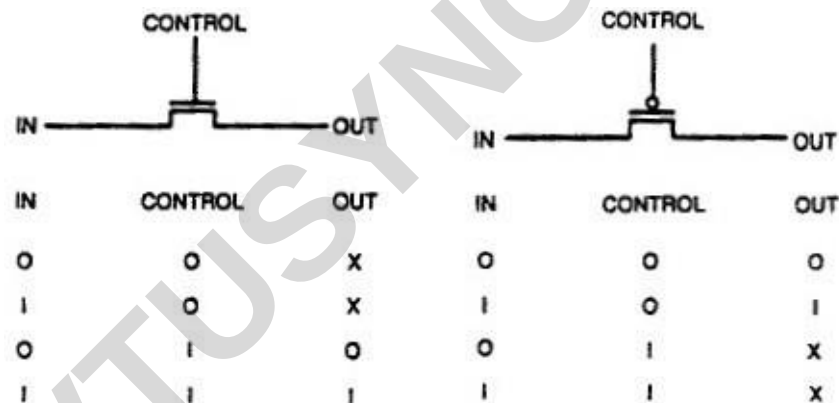


Figure 13: PTL model

8.3.1 XOR Gate using Pass Transistor Logic

Table 2: Truth table of XOR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Let us take A as control signal.

We can observe from the truth table that:

- When $A = 0$; $Y = B$.
- When $A = 1$; $Y = \overline{B}$

The PTL implementation of XOR gate is shown in figure below.

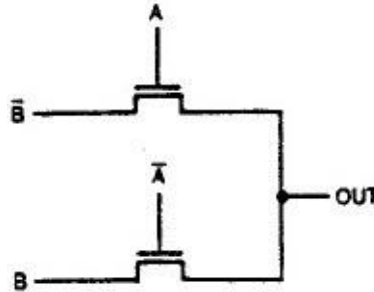


Figure 14: PTL structure for XOR function.

8.4 Challenges and Considerations in Pass Transistor Logic

While PTL offers advantages in reducing transistor count and increasing speed, several challenges must be addressed:

1. Voltage Degradation (V_{th} drop)

The output voltage may not reach the full supply level due to **threshold voltage loss** in nMOS PTL.

Complementary pass networks (nMOS + pMOS) help mitigate this but introduce additional delay.

2. Increased Internal Node Capacitance

The **merging of source and drain regions** makes PTL design complex, leading to **higher internal capacitances**.

3. Requirement for Complementary Control Signals

Both **true and complemented versions** of control variables are needed, increasing circuit complexity.

4. Slower Pull-Down in Complementary Structures

Complementary PTL networks **incur extra delay** in pull-down operations compared to standard CMOS gates.

8.5 Best Practices for Pass Transistor Logic

To achieve efficient PTL circuits, the following techniques should be considered:

Precharged Output Nodes: Helps mitigate voltage degradation issues.

Static PTL with Feedback Buffers: Ensures full voltage swing and eliminates **DC power dissipation**.

Complementary Pass Networks (CVSL): Can be integrated with **CVSL logic** for improved performance.

Finally, any PTL design should be **evaluated through simulation and layout analysis** to determine its effectiveness for a given application.

8.6 Conclusion

Pass Transistor Logic (PTL) provides an efficient alternative to conventional logic gates, particularly in **multiplexers, arithmetic logic units (ALUs), and XOR gates**.

While PTL reduces transistor count and improves speed, it introduces challenges such as **threshold voltage drop, increased capacitance, and control signal complexity**.

By using **complementary networks, precharge techniques, and static implementations**, these challenges can be mitigated, making PTL a valuable approach for optimized circuit design.

9 Electrical and Physical Design of Logic Gates

9.1 The Inverter

9.1.1 Basic Schematic Representation

Figure below illustrates the standard **CMOS inverter circuit**, where:

The **pMOS transistor** is connected to the power supply (V_{DD}).

The **nMOS transistor** is connected to ground (V_{SS}).

The input is applied to both transistor gates.

The output is taken from the connection between the pMOS drain and nMOS drain.

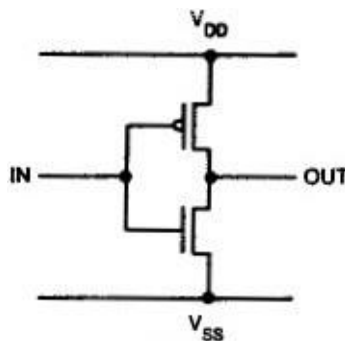


Figure 15: CMOS Inverter Circuit

In a schematic, connections are represented by lines. However, in a physical layout, interconnections involve different materials such as **polysilicon, diffusion regions, and metal layers**.

9.1.2 Physical Layout Considerations

To implement the inverter physically, we must account for:

The **nMOS source and drain** being in **n-diffusion**.

The **pMOS source and drain** being in **p-diffusion**.

The fact that direct **n-diffusion to p-diffusion** connections are not possible in bulk CMOS.

The need for metal interconnects to connect different layers.

Figure below shows the symbolic layout representation where:

The **inverter's drains** are connected using a **metal wire and two contacts**.

Power (V_{DD}) and ground (V_{SS}) connections use metal for low resistance.

The **gate connection** is made using a **polysilicon wire**.

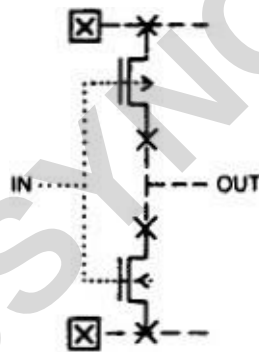


Figure 16: Symbolic layout representation of CMOS Inverter

The corresponding layout is shown in figure below.

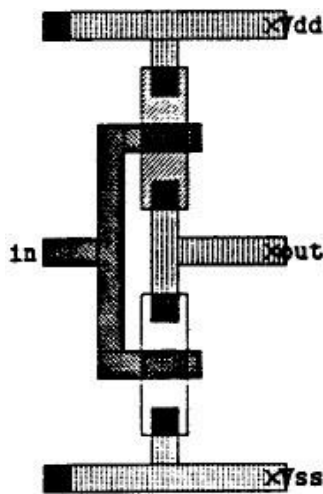


Figure 17: Standard Layout of CMOS Inverter

An alternative layout is shown in figure below where the transistors are aligned horizontally.

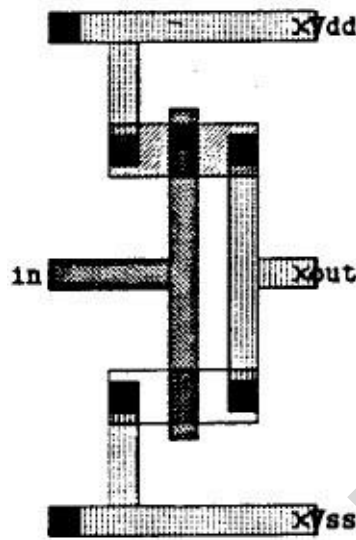


Figure 18: Layout of CMOS Inverter with horizontal transistors

9.1.3 Alternative Layout Strategies

Topology Variations Different topologies exist for inverter layouts:

1. Horizontal Metal Straps with Vertical Polysilicon:

Suitable when a **metal line must pass through the cell**.

The **vertical polysilicon** connects the transistor drains.

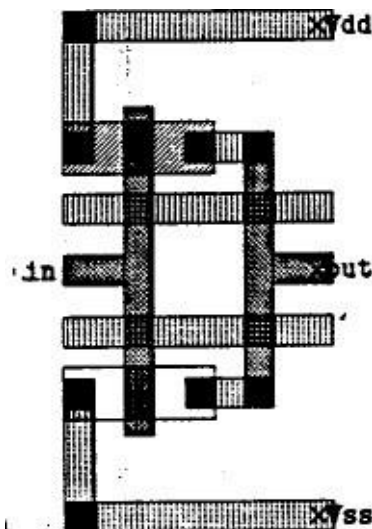


Figure 19: Alternative CMOS inverter layout with vertical polysilicon

2. Power and Ground in Diffusion Layers:

Enables **horizontal metal interconnections** through the cell.
 Reduces metal congestion.

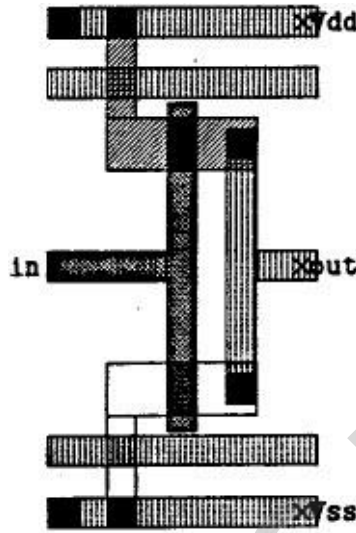


Figure 20: CMOS inverter layout with diffusion power and ground

Performance Considerations

Using a **vertical polysilicon drain connection** introduces extra **resistance and capacitance**:

$$R_{total} \approx 2R_{contact} + R_{poly}$$

where $R_{contact}$ is the resistance of a metal-polysilicon contact, and R_{poly} is the resistance of the polysilicon wire.

Using **diffusion for power and ground connections** adds **series resistance and capacitance**.

To ensure good performance, the resistance should be at least **an order of magnitude lower** than the transistor's "on" resistance.

Using Additional Metal Layers

The introduction of a **second metal layer** enhances layout flexibility:

The **second metal layer** can be used for V_{DD} and V_{SS} supply lines.

Alternatively, it can be used to **strap polysilicon** to reduce resistance and improve signal propagation.

Layouts remain largely unchanged except for the addition of **metal-2 wires** and **metal-1 connection stubs**.

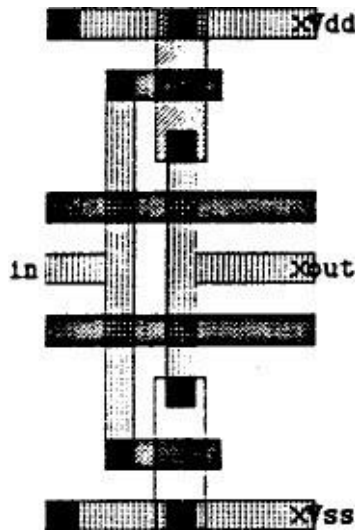


Figure 21: CMOS inverter layout using additional metal layers

9.1.4 Large Inverter Designs

A large inverter can be created by **parallelizing smaller inverters**.

Source and drain regions are merged to reduce resistance.

The diffusion regions are stitched together to minimize **parasitic capacitance**.

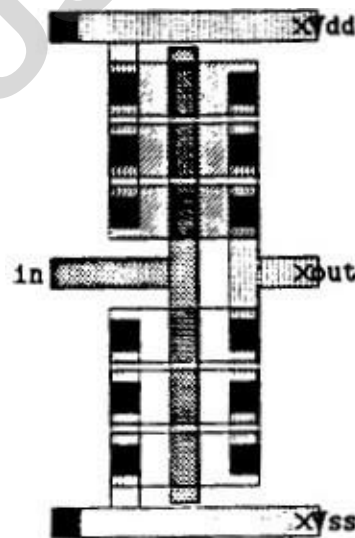


Figure 22: Large CMOS inverter using parallel transistors

Further improvements can be achieved by placing transistors **back-to-back** to optimize **drain capacitance**.

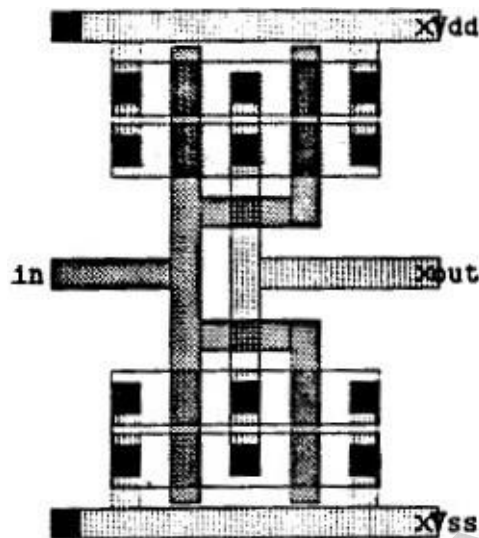


Figure 23: Back-to-Back transistor placement for large inverters

9.1.5 Advanced Drain Capacitance Reduction

Further drain capacitance reduction can be achieved using the **star connection**.

The transistors' drains form a **single continuous region**.

No corner gaps exist, which increases **transistor gain** (β) while minimizing capacitance.

The **effective gain is quadrupled** with minimal drain area increase.

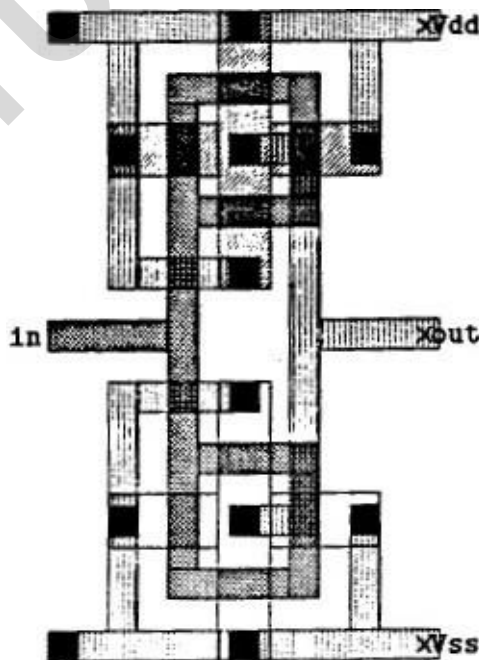


Figure 24: Star connection for CMOS inverter layout optimization

9.2 NAND Gate

The design principles used for the CMOS inverter layout can be extended to implement NAND and NOR gates.

The layout considerations involve transistor orientation, interconnect optimization, and capacitance minimization.

9.2.1 NAND Gate Layout

Basic Layout Translation

A 2-input NAND gate can be implemented using a combination of series nMOS transistors and parallel pMOS transistors.

Figure below represents the direct translation of schematic into layout of a 2-input NAND gate.

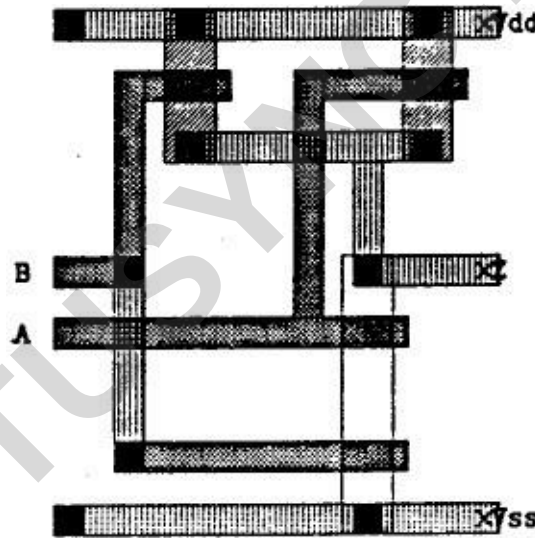


Figure 25: Basic layout translation of a 2-input NAND gate

Horizontal Transistor Orientation

By orienting the transistors horizontally, we obtain the layout shown in figure below, which is cleaner and more compact.

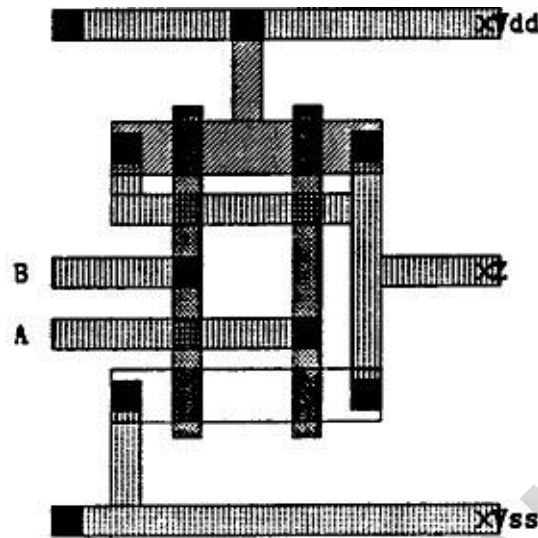


Figure 26: Optimized layout for NAND Gate

9.2.2 Design Style

For multiple-input static gates, the following layout style is adopted:

Transistors are oriented horizontally.

Polysilicon gate signals run vertically.

In cases where deviations from this style occur, specific design reasons will be provided.

The NAND gate could be rotated by 90° to have vertical metal and horizontal polysilicon connections.

9.3 NOR Gate

9.3.1 NOR Gate Layout

Symbolic Layout A 2-input NOR gate follows the opposite arrangement of the NAND gate, where:

nMOS transistors are in parallel.

pMOS transistors are in series.

The symbolic layout of the NOR gate is shown in figure below.

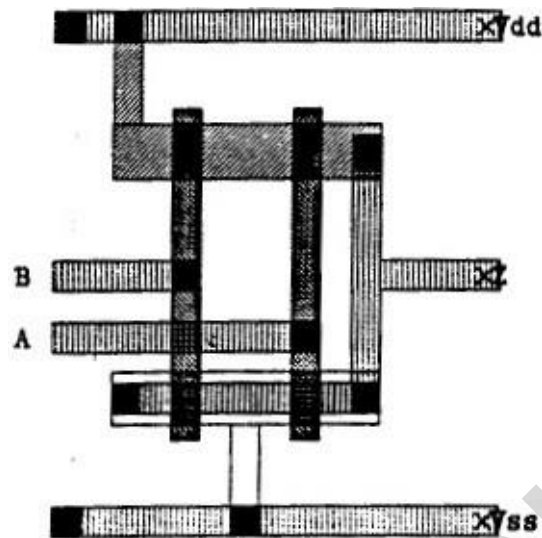


Figure 27: Symbolic layout of NOR gate

Alternative Layout for Faster Performance

An alternative NOR layout is shown in figure below, where the connection to the parallel transistors is optimized.

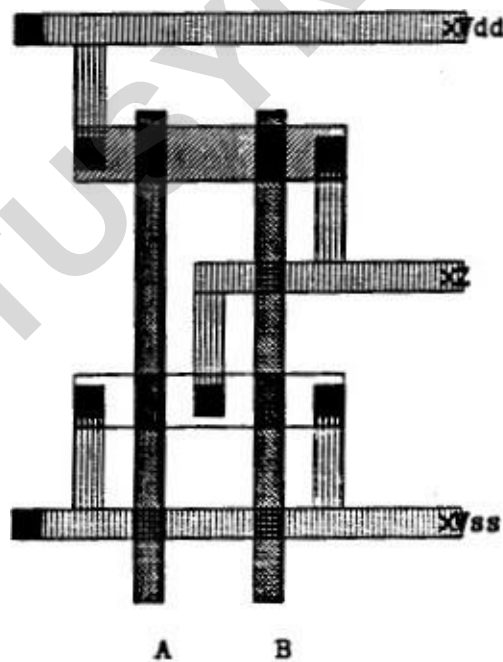


Figure 28: An alternative layout of NOR gate

The alternative layout has the following advantages:

- Reduced drain area connected to the output, minimizing capacitance.

- Faster gate operation, since the reduced capacitance improves switching speed.

The same optimization can be applied to the NAND gate, improving its speed.

9.4 Factors Affecting Complex Gate Design

For more complex logic gates, four key factors influence their electrical and physical design:

1. **Series Transistor Connection:** Increased resistance affects speed.
2. **Body Effect:** Threshold voltage variation impacts performance.
3. **Source-Drain Capacitance:** Larger diffusion areas increase capacitance.
4. **Charge Redistribution:** Affects node voltages and dynamic behavior.

9.5 Body Effect

The **body effect** refers to the modification of the **threshold voltage** (V_t) due to a voltage difference between the source and substrate. Specifically, the threshold voltage variation can be expressed as:

$$\Delta V_t = \gamma \sqrt{V_{sb}}$$

where:

ΔV_t is the change in threshold voltage,

V_{sb} is the source-to-body voltage,

The proportionality constant (γ) depends on the fabrication process.

9.5.1 Impact on NAND Gates

In a multi-input NAND gate, the nMOS transistor at the output switches slower if its source potential is not the same as the substrate.

Figure below illustrates such a scenario.

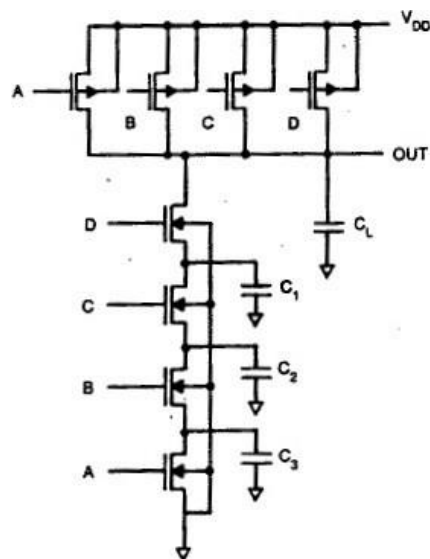


Figure 29: Multiple-input NAND gate affected by body effect

Consider the situation in figure below:

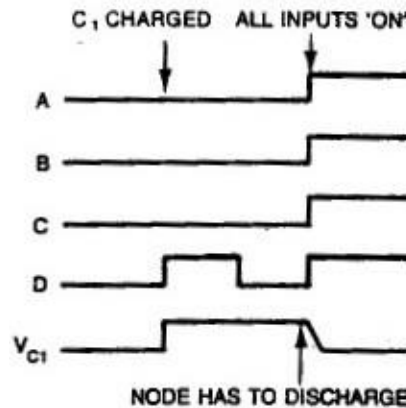


Figure 30: Capacitive charging effect in body effect

Initially, nMOS transistors A, B, and C are OFF ($V_{gsA} = V_{gsB} = V_{gsC} = 0$).

The nMOS transistor D is turned ON, causing $V_{gsD} = V_{DD}$.

When transistor D is switched OFF, the internal node capacitance C_1 gets charged, leading to a nonzero V_{sb} .

If all inputs are then driven HIGH ($V_{gsA} = V_{gsB} = V_{gsC} = V_{DD}$), the source of transistor D will be at $V_{DD} - V_t$.

The fall time of the gate output will be slower than expected due to this internal charge buildup.

9.5.2 Minimizing the Body Effect

To reduce body effect and improve circuit speed, the following strategies should be used:

Minimize internal node capacitance to reduce parasitic charging effects.

Consider the relative impact of body effect in nMOS and pMOS transistors.

If nMOS body effect is worse, using NOR structures instead of NAND may be preferable.

9.5.3 Optimization Techniques

Since body effect is a dynamic issue related to parasitic capacitances, two optimization techniques can be applied:

1. Optimized Transistor Placement:

Place transistors with late-arriving signals closest to the output.

Early-arriving signals discharge internal nodes, reducing body effect for transistors switching later.

2. Minimizing Internal Node Capacitance:

Use metal interconnects instead of diffusion wires for internal node connections.

If buried contacts are allowed, use polysilicon instead of diffusion for interconnects.

Minimize the diffusion area attached to transistors to reduce parasitic capacitance.

9.6 Physical Layout of Logic gates

All complementary CMOS gates can be designed using a single row of nMOS transistors and a single row of pMOS transistors, aligned at common gate connections.

Most simple gates can be designed using an unbroken row of transistors, following the line of diffusion rule.

This means that source-drain connections are made by abutment.

9.6.1 Graph Representation of CMOS Circuits

To automate layout generation, a graph-theoretic approach is used.

The CMOS circuit is converted into a graph, where:

1. Vertices represent the source/drain connections.
2. Edges represent transistors that connect particular source-drain vertices.

This results in two graphs: one for the nMOS network and another for the pMOS network.

An example of this graph transformation is shown in figure below.

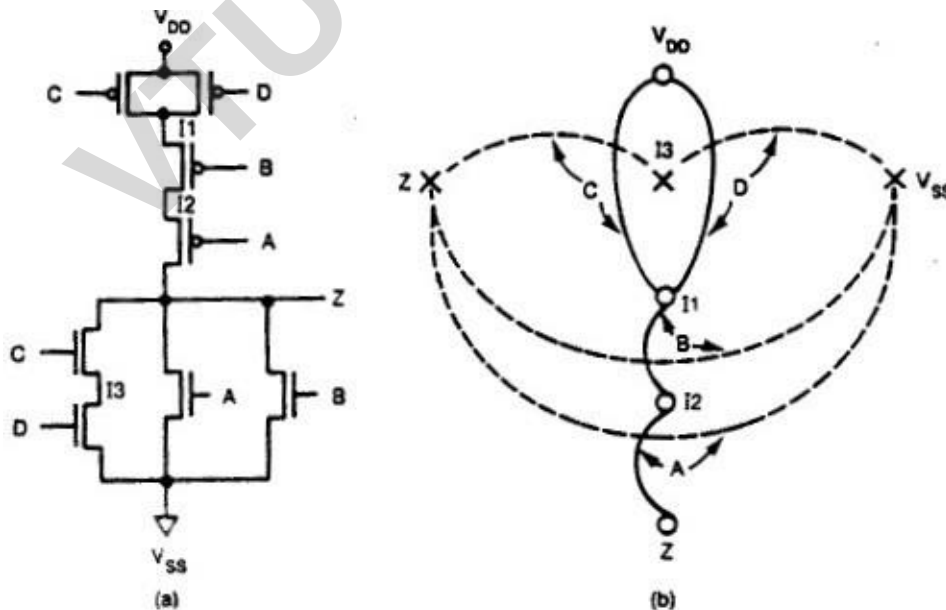


Figure 31: CMOS logic gate graph representation

The p-graph and n-graph are dual to each other, as the pMOS pull-up network is the complement of the nMOS pull-down network.

9.6.2 Euler Paths in CMOS Layout

If two edges are adjacent in the p- or n-graph, then:

They may share a common source-drain connection.

They can be connected by abutment.

A gate can be designed through breaks if an Euler path exists in both the p-graph and n-graph with identical labeling. The main algorithm for finding such paths is:

1. Find all Euler paths in the graph.
2. Identify a p- and n-Euler path with identical labeling.
3. If none exist, break the gate in the minimum number of places to achieve this condition.

An Euler path example and the corresponding layout generation is shown in figure below.

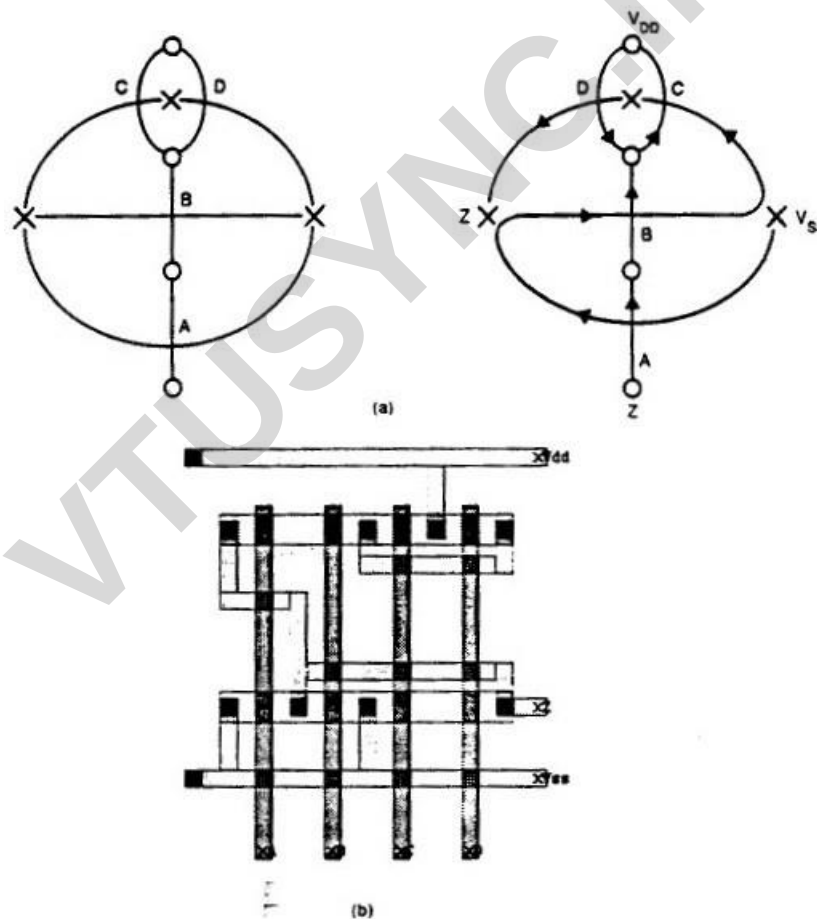


Figure 32: Euler paths in CMOS gate and the corresponding layout

9.6.3 Automated Layout Techniques

A graph-theoretic approach can automate layout generation, using interval graphs to optimally place transistors in a gate matrix style.

The final layout is shown in figure below.

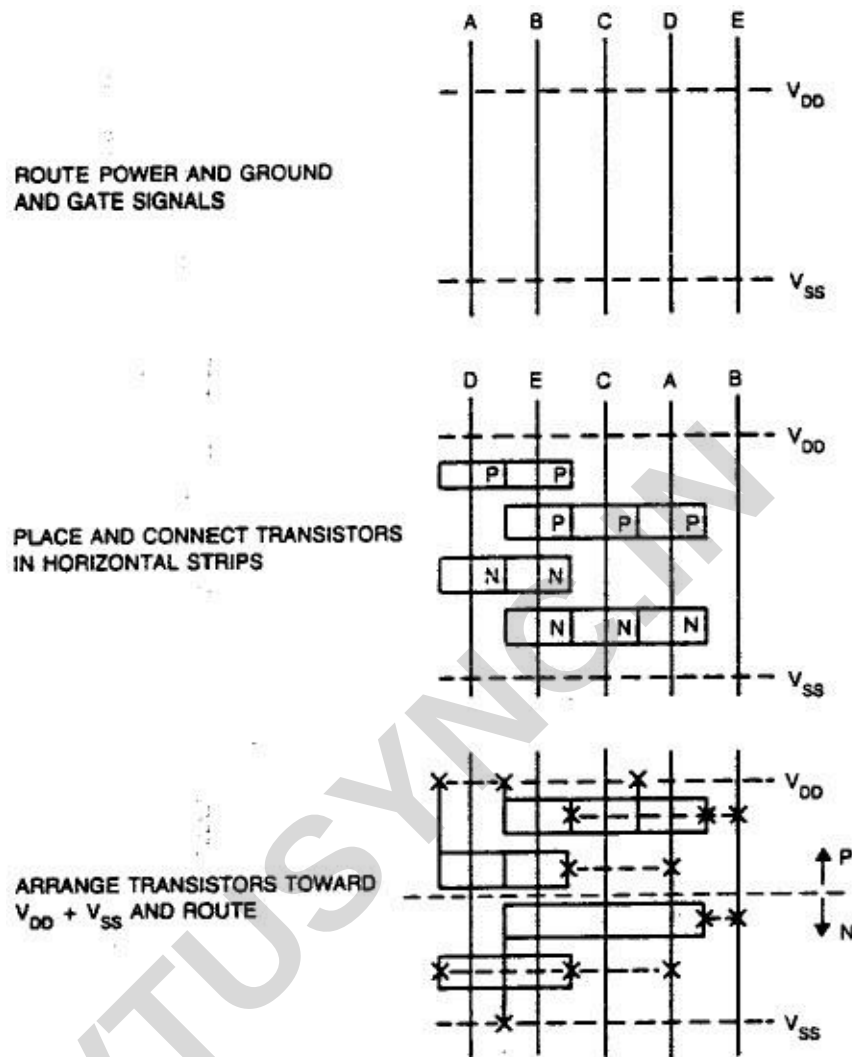


Figure 33: Outline of automated approach to CMOS gate layout

Steps in this approach:

1. Group transistors in strips to maximize source/drain abutment.
2. Interchange polysilicon columns to improve abutment.
3. Place transistor groups in rows, with V_{DD} and GND rails at the edges.
4. Route connections using metal or diffusion wires.

10 Input-output (I/O) structures

CMOS I/O structures are critical in VLSI due to their sensitivity to both circuit design and process-specific characteristics.

In practice, system designers should use well-characterized **library cells** instead of designing pads from scratch.

These notes provide foundational insights into the structure and layout of I/O pads.

10.1 Overall Organization of I/O Pads

Standard pad size: $\sim 150\mu\text{m} \times 150\mu\text{m}$, suitable for bond wire attachment.

Connection points and V_{DD}/V_{SS} rails are placed consistently for uniformity.

Wider power/ground buses are used based on worst-case power estimates; multiple pads may be employed to reduce noise.

Often, V_{SS} is placed as the outermost rail for protection.

Figure below illustrate some of these concepts.

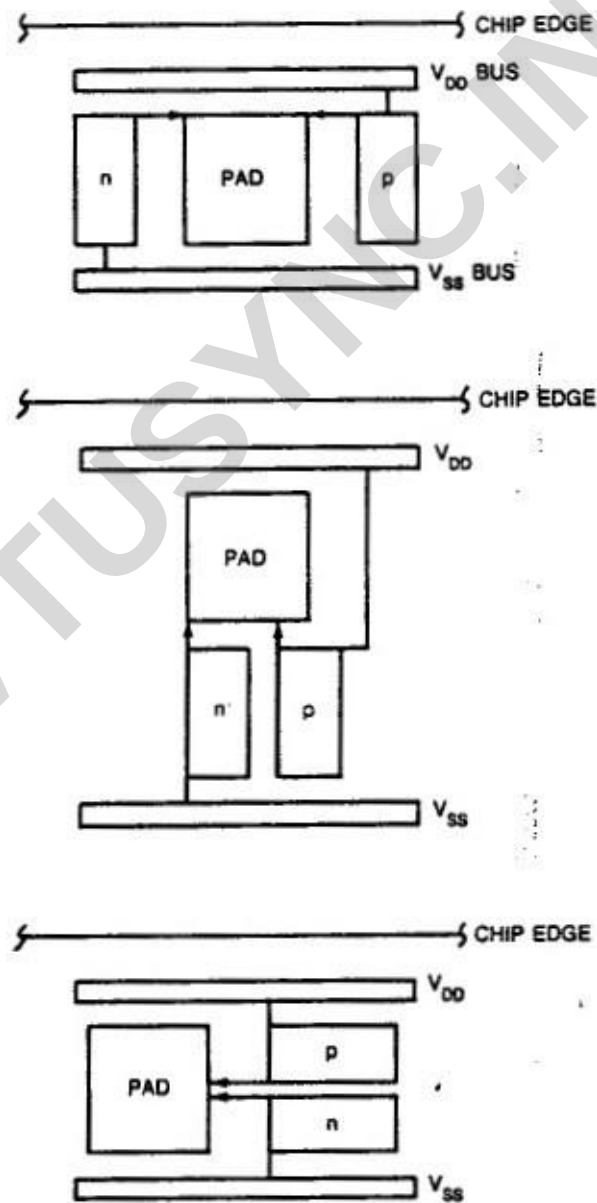


Figure 34: General pad layouts

Pad frames may be auto-generated from text descriptions. Example:

```

LEFT;
  INPUT A
  INPUT B
TOP;
  VDD VDD
  INPUT C
RIGHT;
  OUTPUT Z
  OUTPUT Y
BOTTOM;
  OUTPUT W
  VSS VSS
  
```

The resulting I/O frame is shown in figure below.

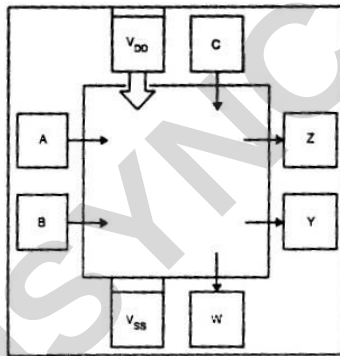


Figure 35: I/O frame generation

10.2 VDD and VSS Pads

Simple metal pads connected directly to power rails.

If discontinuity occurs, complete the path using:

- Polysilicon bridges as shown in figure below.

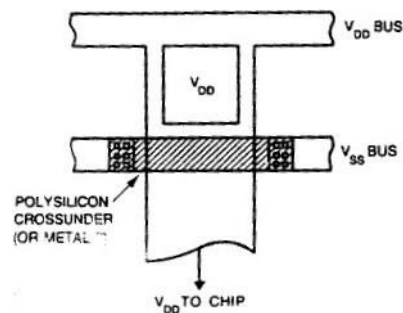


Figure 36: VDD pad design

- Multi-metal crossover with sufficient vias

10.3 Output Pads

Key Requirements

Sufficient drive strength to drive capacitive loads.

Adequate rise/fall times.

DC characteristics compliance when driving TTL or other logic levels.

Design Strategy

Use two-stage inverters with optimal sizing (2.7:1 or within 2–10 range).

Include buffering to reduce internal loading.

Apply latch-up protection:

- Separate nMOS/pMOS regions
- Use guard rings tied to VDD/VSS

TTL Load Compatibility

CMOS $V_{OH} = 5V$, TTL $V_{IH} = 2.4V \Rightarrow OK$

CMOS $V_{OL} = 0V$, TTL $V_{IL} = 0.4V \Rightarrow OK$

Must sink 1.6 mA for LOW state $\Rightarrow$ usually not an issue

10.4 Input Pads

The design of input pads shares several principles with output pad design, particularly in terms of transistor sizing.

In fact, transistors used in output pads can often be reused by simply reversing their function.

However, **one critical issue** must be addressed in input pad design: **electrostatic discharge (ESD)** protection.

High Input Resistance and Gate Breakdown

The gate of a MOS transistor typically exhibits **very high input resistance**, in the range of 10^{12} to $10^{13} \Omega$.

As a result, even tiny leakage currents can cause a substantial voltage build-up across the gate oxide.

The voltage V that develops on the gate can be estimated using the equation:

$$V = \frac{I \Delta t}{C_g}$$

where:

- I is the charging current,
- Δt is the charging time, and
- C_g is the gate capacitance.

Example: Let:

$$I = 10 \mu\text{A}, \quad C_g = 0.03 \text{ pF}, \quad \Delta t = 1 \mu\text{s}$$

Then:

$$V = \frac{10 \times 10^{-6} \times 1 \times 10^{-6}}{0.03 \times 10^{-12}} \approx 330 \text{ V}$$

Such high voltages can easily **exceed the oxide breakdown voltage** (typically 40 - 100 V), causing permanent damage.

ESD Protection: Clamp Diodes and Series Resistance

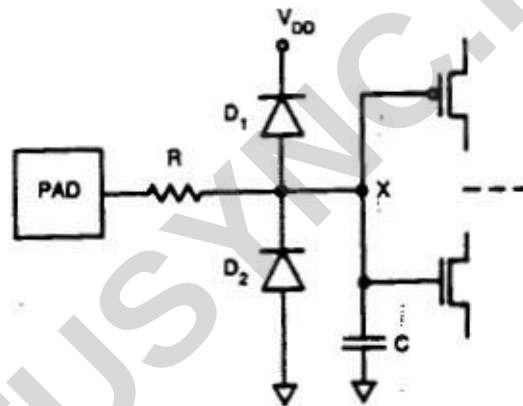


Figure 37: Typical input protection circuit

To protect the gate from high-voltage transients, **input pads include a resistor and diode clamps**:

- **Clamp Diodes (D1 and D2):** Turn on if the input voltage exceeds V_{DD} or falls below V_{SS} .
- **Series Resistor (R):** Limits the peak current during voltage excursions. Typical values: 200Ω to $3k\Omega$.

Note: The resistor and the parasitic input capacitance form an RC time constant, which can limit high-speed operation.

A **polysilicon resistor** is preferable over a diffusion resistor in a **p-well process**, as it reduces unwanted substrate charge injection and helps avoid latch-up.

In an **n-well process**, full **n-device I/O circuitry** can be built.

Here, n^+ diffused protection resistors and **punch-through devices** are employed:

- A **punch-through device** has closely spaced source and drain regions but no gate.
- It acts like an avalanche diode, turning on around 50 V.
- This structure does not require additional well formation.

Interfacing TTL Logic with CMOS

When interfacing TTL outputs with CMOS inputs:

TTL levels: $V_{OL} = 0.4\text{ V}$, $V_{OH} = 2.4\text{ V}$

CMOS inverters should switch around **1.4 V** (midpoint of TTL range)

This can be achieved either by:

Ratioing the p- and n-MOS sizes in the inverter, or

Using a **reference voltage**.

In some cases, the TTL output may use an **external resistor to 5 V** to improve V_{OH} . This resistor can even be integrated into the pad using a **p-MOS transistor**.

10.5 Tri-state Pads

Based on tri-state inverter design.

Alternate design, as shown in figure below, offers faster switching.

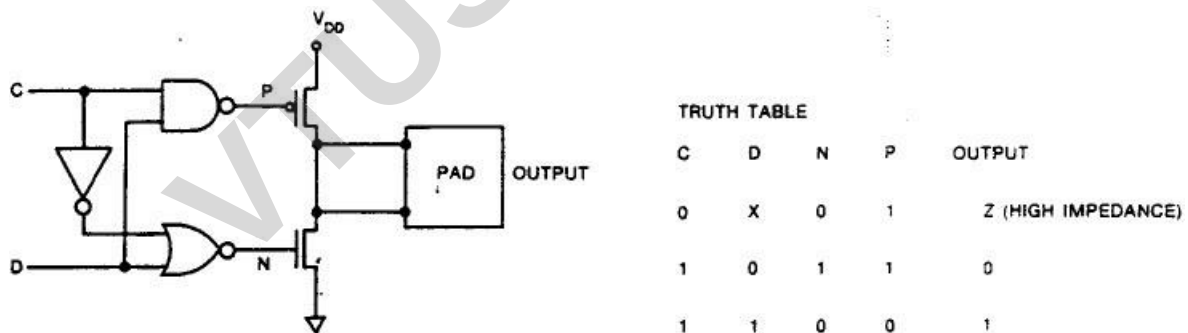


Figure 38: Tri-state Pad

Must prevent short-circuit (DC) current during transitions.

10.6 Bi-directional Pads

Combine input pad + tri-state output buffer.

Example shown in figure below.

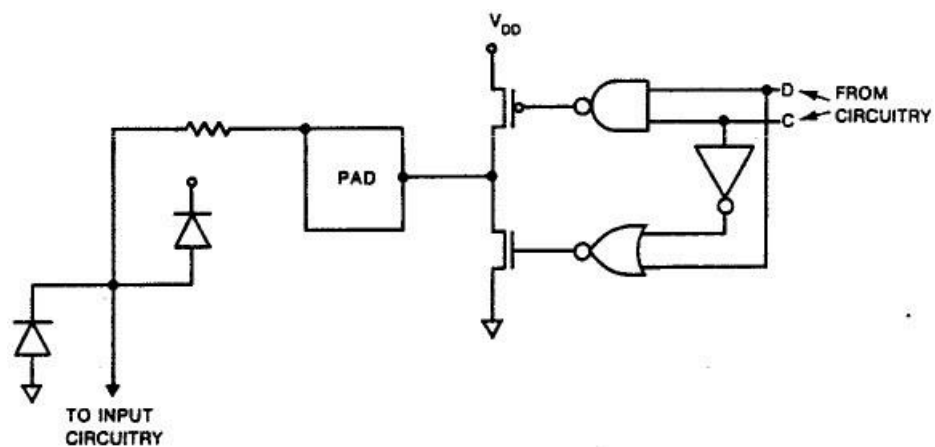


Figure 39: Bi-directional Pad

Useful for shared data/address lines.